

- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy
- Thiothixene should generally not be used during the first trimester
- Thiothixene should be used during pregnancy only if clearly needed

Breast Feeding

- Unknown if thiothixene is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The art of psychopharmacology

Potential Advantages

- For patients who do not respond to other antipsychotics

Potential Disadvantages

- Patients with tardive dyskinesia
- Children
- Elderly

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis

Pearls

- * Although not systematically studied, may cause less weight gain than other antipsychotics
- Conventional antipsychotics are less expensive than atypical antipsychotics
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another
- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional antipsychotic such as thiothixene or from switching to a conventional antipsychotic such as thiothixene
- However, long-term polypharmacy with a combination of a conventional antipsychotic such as thiothixene with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2

atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring

- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

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Tiagabine

Therapeutics

Brands

- Gabitril
- *see index for additional brand names*

Generic?

Yes

Class

- Anticonvulsant; selective GABA reuptake inhibitor (SGRI)

Commonly Prescribed for

(bold for FDA approved)

- **Partial seizures (adjunctive; adults and children 12 years and older)**
- Anxiety disorders
- Neuropathic pain/chronic pain

How the Drug Works

- Selectively blocks reuptake of gamma-aminobutyric acid (GABA) by presynaptic and glial GABA transporters

How Long Until It Works

- Should reduce seizures by 2 weeks
- Not clear that it works in anxiety disorders or chronic pain but some patients may respond, and if they do, therapeutic actions can be seen by 2 weeks

If It Works

- The goal of treatment is complete remission of symptoms (e.g., seizures, anxiety)
- The goal of treatment of chronic neuropathic pain is to reduce symptoms as much as possible, especially in combination with other treatments
- Treatment of chronic neuropathic pain most often reduces but does not eliminate symptoms and is not a cure since symptoms usually recur after medicine stopped
- Continue treatment until all symptoms are gone or until improvement is stable and then continue treating indefinitely as long as improvement persists

If It Doesn't Work (for Neuropathic Pain or Anxiety Disorders)

- Many patients have only a partial response where some symptoms are improved but others persist
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- May only be effective in a subset of patients with neuropathic pain or anxiety disorders, in some patients who fail to respond to other treatments, or it may not work at all
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider biofeedback or hypnosis for pain
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Switch to another agent with fewer side effects



Best Augmenting Combos for Partial Response or Treatment Resistance

- Tiagabine is itself an augmenting agent for numerous other anticonvulsants in treating epilepsy
- * For neuropathic pain, tiagabine can augment TCAs and SNRIs as well as gabapentin, other anticonvulsants, and even opiates if done by experts while carefully monitoring in difficult cases
- For anxiety, tiagabine is a second-line treatment to augment SSRIs, SNRIs, or benzodiazepines

Tests

- None for healthy individuals
- Tiagabine may bind to tissue that contains melanin, so for long-term treatment ophthalmological checks may be considered

Side Effects

How Drug Causes Side Effects

- CNS side effects may be due to excessive actions of GABA

Notable Side Effects

- * Sedation, dizziness, asthenia, nervousness, difficulty concentrating, speech/language problems, confusion, tremor
- Diarrhea, vomiting, nausea
- Ecchymosis, depression



Life-Threatening or Dangerous Side Effects

- Exacerbation of EEG abnormalities in epilepsy
- Status epilepticus in epilepsy (unknown if associated with tiagabine use)
- Sudden unexplained deaths have occurred in epilepsy (unknown if related to tiagabine use)

- New onset seizures and status epilepticus have been reported in patients without epilepsy
- Rare activation of suicidal ideation and behavior (suicidality)

Weight Gain



- Reported but not expected
- Some patients experience increased appetite

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take more of the dose at night or all of the dose at night to reduce daytime sedation
- Lower the dose
- Switch to another agent

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 32–56 mg/day in 2–4 divided doses for adjunctive treatment of epilepsy
- 2–12 mg/day for adjunctive treatment of chronic pain and anxiety disorders

Dosage Forms

- Tablet 2 mg, 4 mg, 12 mg, 16 mg, 20 mg

How To Dose

- Adjunct to enzyme-inducing antiepileptic drugs: initial 4 mg once daily; after 1 week can increase dose by 4–8 mg/day each week; maximum dose generally 56 mg/day in 2–4 divided doses
- Dosing for chronic pain or anxiety disorders not well established, but start as low as 2 mg at night, increasing by 2 mg increments every few days as tolerated to 8–12 mg/day
- Exercise particular caution when prescribing in uninduced patients



Dosing Tips

- Usually administered as adjunctive medication to other anticonvulsants in the treatment of epilepsy
- * Dosing recommendations are based on studies of adjunctive use with enzyme-inducing antiepileptic drugs, which lower plasma levels of tiagabine by half; thus, when tiagabine is used without enzyme-inducing antiepileptic drugs the dose may need to be significantly reduced and may require a much slower titration rate
- * Also administered as adjunctive medication to benzodiazepines, SSRIs, and/or SNRIs in the treatment of anxiety disorders; and to SNRIs, gabapentin, other anticonvulsants, and even opiates in the treatment of chronic pain
- * Dosing varies considerably among individual patients but is definitely at the lower end of the dosing spectrum for patients with chronic neuropathic pain or anxiety disorders (i.e., 2–12 mg either as a split dose or all at night)
- * Patients with chronic neuropathic pain and anxiety disorders are far less tolerant of CNS side effects, so they require a much slower dosage titration as well as a lower maintenance dose
- Gastrointestinal absorption is markedly slowed by the concomitant intake of food, which also lessens the peak plasma concentrations
- * Thus, for improved tolerability and consistent clinical actions, instruct patients to always take with food
- Side effects may increase with dose

Overdose

- No fatalities have been reported; sedation, agitation, confusion, speech difficulty, hostility, depression, weakness, myoclonus, seizures, status epilepticus

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper
- Epilepsy patients may seize upon withdrawal, especially if withdrawal is abrupt
- Discontinuation symptoms uncommon

Pharmacokinetics

- Primarily metabolized by CYP450 3A4
- Steady-state concentrations tend to be lower in the evening than in the morning
- Half-life approximately 7–9 hours
- Renally excreted

Drug Interactions

- Clearance of tiagabine may be reduced and thus plasma levels increased if taken with a non-enzyme-inducing antiepileptic drug (e.g., valproate, gabapentin, lamotrigine), so tiagabine dose may need to be reduced
- CYP450 3A4 inducers such as carbamazepine can lower the plasma levels of tiagabine
- CYP450 3A4 inhibitors such as nefazodone, fluvoxamine, and fluoxetine could theoretically increase the plasma levels of tiagabine
- Clearance of tiagabine is increased if taken with an enzyme-inducing antiepileptic drug (e.g., carbamazepine, phenobarbital, phenytoin, primidone) and thus plasma levels are reduced ; however, no dose adjustments are necessary for treatment of epilepsy as the dosing recommendations for epilepsy are based on adjunctive treatment with an enzyme-inducing antiepileptic drug
- Despite common actions upon GABA, no pharmacodynamic or pharmacokinetic interactions have been shown when tiagabine is combined with the benzodiazepine triazolam or with alcohol
- However, sedating actions of any two sedative drugs given in combination can be additive

Other Warnings/Precautions

- Seizures have occurred in individuals without epilepsy who took tiagabine

- Risk of seizure may be dose-related; when tiagabine is used in the absence of enzyme-inducing antiepileptic drugs, which lower plasma levels of tiagabine, the dose may need to be reduced
- Depressive effects may be increased by other CNS depressants (alcohol, MAOIs, other anticonvulsants, etc.)
- Tiagabine may bind to melanin, raising the possibility of long-term ophthalmologic effects
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

Do Not Use

- If there is a proven allergy to tiagabine

Special Populations

Renal Impairment

- Although tiagabine is renally excreted, the pharmacokinetics of tiagabine in healthy patients and in those with impaired renal function are similar and no dose adjustment is recommended

Hepatic Impairment

- Clearance is decreased
- May require lower dose

Cardiac Impairment

- No dose adjustment recommended

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy not established in children under age 12
- Maximum recommended dose generally 32 mg/day in 2–4 divided doses



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, www.aedpregnancyregistry.org
- Taper drug if discontinuing

- Seizures, even mild seizures, may cause harm to the embryo/fetus
- * Lack of definitive evidence of efficacy for chronic neuropathic pain or anxiety disorders suggests risk/benefit ratio is in favor of discontinuing tiagabine during pregnancy for those indications

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects
- If infant shows signs of irritability or sedation, drug may need to be discontinued

The Art of Psychopharmacology

Potential Advantages

- Treatment-resistant chronic neuropathic pain
- Treatment-resistant anxiety disorders

Potential Disadvantages

- May require 2–4 times a day dosing
- Needs to be taken with food

Primary Target Symptoms

- Incidence of seizures
- Pain
- Anxiety

Pearls

- Well studied in epilepsy
- Much use is off label
- * Off-label use second-line and as an augmenting agent may be justified for treatment-resistant anxiety disorders and neuropathic pain and also for fibromyalgia
- * Off-label use for bipolar disorder may not be justified
- * One of the few agents that enhances slow-wave delta sleep, which may be helpful in chronic neuropathic pain syndromes
- Can be difficult to dose in patients who are not taking enzyme-inducing anticonvulsant drugs as the doses in uninduced patients have not been well studied, are generally much lower, and titration is much slower than in induced patients
- Can cause seizures even in patients without epilepsy, especially in patients taking other agents (antidepressants, antipsychotics, stimulants, narcotics) that are thought to lower the seizure threshold

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Tianeptine

Therapeutics

Brands

- Coaxil
- Stablon
- Tatinol
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: glutamate; yet to be determined (Glu)
- Glutamatergic modulator
- Often classified as a tricyclic antidepressant, but pharmacologically distinct

Commonly Prescribed for

(bold for FDA approved)

- Major depressive disorder

- Dysthymia
- Anxiety associated with depression

How the Drug Works

- * Modulates glutamatergic neurotransmission, perhaps through potentiation of AMPA (alpha-amino-3- hydroxy-5-methyl-4-isoxazolepropionic acid) receptor function
- Full agonist at mu opioid receptors; animal studies suggest that this may contribute to its antidepressant effects

How Long Until It Works

- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)

- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)

Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation has not been systematically studied with tianeptine

Tests

- None recommended for healthy individuals

Side Effects

How Drug Causes Side Effects

- * Mild anticholinergic activity (less than some TCAs) could possibly lead to sedative effects, dry mouth, constipation, and blurred vision
- Most side effects are immediate but often go away with time
- * Pharmacologic studies do not indicate tianeptine to be a potent alpha 1 antagonist or H1 antihistamine

Notable Side Effects

- Headache, dizziness, insomnia, sedation
- Nausea, constipation, abdominal pain, dry mouth
- Abnormal dreams
- Increased transaminases
- Tachycardia



Life-Threatening or Dangerous Side Effects

- Theoretically, rare induction of mania and activation of suicidal ideation or behavior
- Cases of activation of suicidal ideation and behavior (suicidality) (short-term did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)
- Hepatitis that can, in exceptional cases, be severe

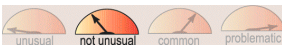
- Dermatitis bulbous in exceptional cases

Weight Gain



- Not well studied

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- In a few weeks, switch or add other drugs
- For skin reactions, stop treatment

Best Augmenting Agents for Side Effects

- Augmentation for side effects of tianeptine has not been systematically studied

Dosing and Use

Usual Dosage Range

- 37.5 mg/day

Dosage Forms

- Tablet 12.5 mg

How to Dose

- 12.5 mg 3 times/day



Dosing Tips

- Tianeptine's rapid elimination necessitates strict adherence to the dosing schedule
- * Short half-life means multiple daily doses

Overdose

- Effects are generally mild and nonfatal; unlikely to cause cardiovascular effects

Long-Term Use

- Safe

Habit Forming

- Abuse and dependence may occur, in particular in patients under 50 years of age with a history of drug or alcohol dependence

How to Stop

- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation
- If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly

Pharmacokinetics

- Not primarily metabolized by CYP 450 enzyme system
- Tianeptine is rapidly eliminated
- Half-life approximately 2.5 hours



Drug Interactions

- Activation and agitation, especially following switching or adding antidepressants, may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of tianeptine
- Other drug interactions not well studied



Other Warnings/Precautions

- For elective surgery, tianeptine should be stopped 24–48 hours before general anesthesia is administered

- Generally, use only with extreme caution with MAOIs; do not use until 14 days after MAOIs are stopped; do not start an MAOI for at least 5 half-lives (5 to 7 days for most drugs) after discontinuing tianeptine
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents
- Warn doctors to pay attention to patients with history of drug dependencies

Do Not Use

- If patient is taking an MAOI
- If patient is pregnant or nursing
- If there is a proven allergy to tianeptine

Special Populations

Renal Impairment

- Dose should be reduced for severe impairment to 25 mg/day

Hepatic Impairment